Duc Hoang

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Summary

PhD candidate in High-Energy Physics at MIT and CERN developing real-time machine learning systems for nanosecond-scale decision making. My work spans FPGA-optimized neural architectures, adaptive hardware-native learning algorithms, and large-scale statistical analyses for precision Higgs measurements.

Education

- **Massachusetts Institute of Technology (MIT)** — *2021 - Present*
  PhD, Physics. Thesis: Harder, Better, Faster, Stronger: Pushing the Data Frontier at the Large Hadon Collider — Cambridge, USA
- **Rhodes College** — *2017-2021*
  B.S., Physics, Summa Cum Laude. Thesis: Triggering on electron-nucleon scattering events at LDMX — Memphis, USA

Professional Experience

- **MIT & European Organization for Nuclear Research (CERN)** — *2021 - Present*
  Graduate Researcher & Next Generation Trigger PhD Fellow — Cambridge, USA & Geneva, Switzerland
  - Coordinated multi-institutional team to develop **ultra-low-latency FPGA-accelerated ML models** for CMS Level-1 Trigger, which processes approximately 10% of global internet traffic in real time. Hear me explain it along the trigger leardership team **[2]**
  - Built **first FPGA-based ML algorithms for real-time jet tagging** and delivered cross-platform, low-latency data transfer tests between different FPGA board platforms. **[3]**
  - Analyzed all Run 2 CMS data (a few PBs) to deliver **most sensitive Higgs $\rightarrow b\bar{b}$ measurement in the V(qq)H(bb) decay channel**, invited presentation at Rencontres de Moriond 2025. **[4]**
  - Supervised MIT undergraduates and masters on research projects involving the CMS Level-1 Trigger system and the development of **novel machine learning architectures for ultra-fast and efficient inference on FPGAs. [1,9]**.
  - Designed and led practical CERN training sessions on machine learning techniques for FPGA-based Level-1 Trigger applications for CERN students and researchers.
  - Developed data analysis projects for MIT 8.16/8.316: Data Science in Physics, which is published on **MITx** course **Computational Data Science in Physics**.
- **Espresso (MIT Sandbox Team)** — *2025 – Present*
  Co-founder — Cambridge, USA
  - Developing **efficient real-time and local data processing capabilities** for space and defense industry.
  - Defined the company's technical vision and led early customer discoveries focusing on space applications.
- **Fermi National Accelerator Laboratory** — *2019 - 2021*
  Undergraduate Researcher — Chicago, USA
  - Developer of hls4ml, A package for **ultra-low-latency machine learning inference in FPGAs** using high level synthesis language (HLS). **[5]**
  - Designed superconducting magnet quench detection systems **[6]**
  - Analyzed simulation data to predict potential signal gains for the real-time trigger systems for Light Dark Matter Experiment (LDMX). **[7]**
  - Machine Learning analysis to infer hyper-parameter optimization performance of models used in the MINERvA Experiment **[8]**.

Selected Publications & Software

[1] **D. Hoang** et al. "KANELÉ: Kolmogorov–Arnold Networks for Efficient LUT-based Evaluation," *Proceedings of the 2026 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA'26)*, Seaside, CA, USA, 2026. ***Best Paper Nominee (Top 3 Paper)*** https://arxiv.org/abs/2512.12850

[2] **CMS Collaboration**, "The Phase-2 Upgrade of the CMS Level-1 Trigger", https://cds.cern.ch/record/2714892

[3] **CMS Collaboration**, 2025. "Multiclass object tagging in the CMS Phase-2 Level-1 Correlator Trigger." CMS Detector Performance Summary. https://cds.cern.ch/record/2936315.

[4] **CMS Collaboration**, "Search for a boosted Higgs boson decaying to bottom quark pairs in association with a hadronically decaying W or Z boson with the CMS detector using proton-proton collisions at $\sqrt{s} = 13$ TeV", https://cds.cern.ch/record/2928061

[5] hls4ml **Community**, "hls4ml: An Open-Source Codesign Workflow to Empower Scientific Low-Power Machine Learning Devices"

[6] **D. Hoang** et al., "Intelliquench: an adaptive machine learning system for detection of superconducting magnet quenches.," in *IEEE Transactions on Applied Superconductivity*, doi: 10.1109/TASC.2021.3058229.

[7] **LDMX Collaboration**, "Current Status and Future Prospects for the Light Dark Matter eXperiment" [arXiv:2203.08192[hep-ex]].

[8] **D. Hoang** et al. "Inferring Convolutional Neural Networks' Accuracies from Their Architectural Characterizations," *2019 18th IEEE International Conference On Machine Learning And Applications (ICMLA)*, Boca Raton, FL, USA, 2019, pp. 1388-1391. DOI: 10.1109/ICMLA.2019.00226

[9] **CMS Collaboration**. 2024. "NNPuppiTaus: PUPPI Tau Reconstruction in the Level-1 Trigger with Real-Time Machine Learning." CMS Detector Performance Summary. https://cds.cern.ch/record/2895660.

Mentorship & Outreach

Aarush Gupta, MIT BS/MS Thesis (CS & Physics). *Subsequent position: Jane Street*
Orion Foo, MIT BS (CS & Physics).

- **SEAS - Summer in Engineering and Applied Sciences** *2025*
  Founder & General Manager [🌐]
  - Founded, along with other researchers from Central Vietnam, a STEM summer school, providing under-privileged but highly talented high school students in Central Vietnam the opportunity to study emerging fields such as artificial intelligence, data science, programming, renewable energy, and quantum technology at no cost.

Honors and Awards

- **The 2025 Breakthrough Prize in Fundamental Physics** *2025*
  Received as a member of the CMS Collaboration [🌐]
  - The prize is awarded to physicists from theoretical, mathematical, or experimental physics that have made transformative contributions to fundamental physics, and specifically for recent advances.
- **2025 European Physical Society Conference on High Energy Physics (EPS-HEP) Poster Prize** *2025*
  European Physical Society (EPS) [🌐]
  - Awarded for best poster at EPS-HEP 2025 (out of more than 100 posters), a highly prestigious biannual international conference featuring major results from leading high-energy physics experiments and collaborations.
- **MIT Public Service Fellow (PKG) Fellowship** *2025*
  Priscilla King Gray Public Service Center - MIT [🌐]
  - Awarded to develop and implement a social impact project in collaboration with community partners in Central Vietnam.
- **Next Generation Trigger PhD fellow.** *2024*
  CERN [🌐]
  - Awarded for exceptional PhD students to work along CERN scientists to develop the next generation trigger system at the Large Hadron Collider.
- **2024-25 Christiaan Huygens Graduate Fellowship - MIT.** *2024*
  MIT School of Science
  - Fellowship providing financial support for the full academic year in the School of Science for students with non-traditional backgrounds.
- **MIT-The Royal Game of Ur Champion** *2023*
  MIT Playur II Tournament [🌐]
  - The Royal Game of Ur is the oldest board game in the world.
- **Henry W Kendall (1955) first year graduate fellowship.** *2021*
  MIT Laboratory for Nuclear Science
  - Financial support for first year graduate students.
- **Undergraduate Awards.** *2017-2021*
  Rhodes College
  - Research Award in Physics, Council on Undergraduate Research (CUR) Physics & Astronomy Student Travel Award, W.O Shewmaker Award for Excellence in First-Year Search Program, Award for Excellence in Physics by a First-Year Student, Summa Cum Laude, Sigma Pi Sigma, Phi Beta Kappa, Physics Honors.